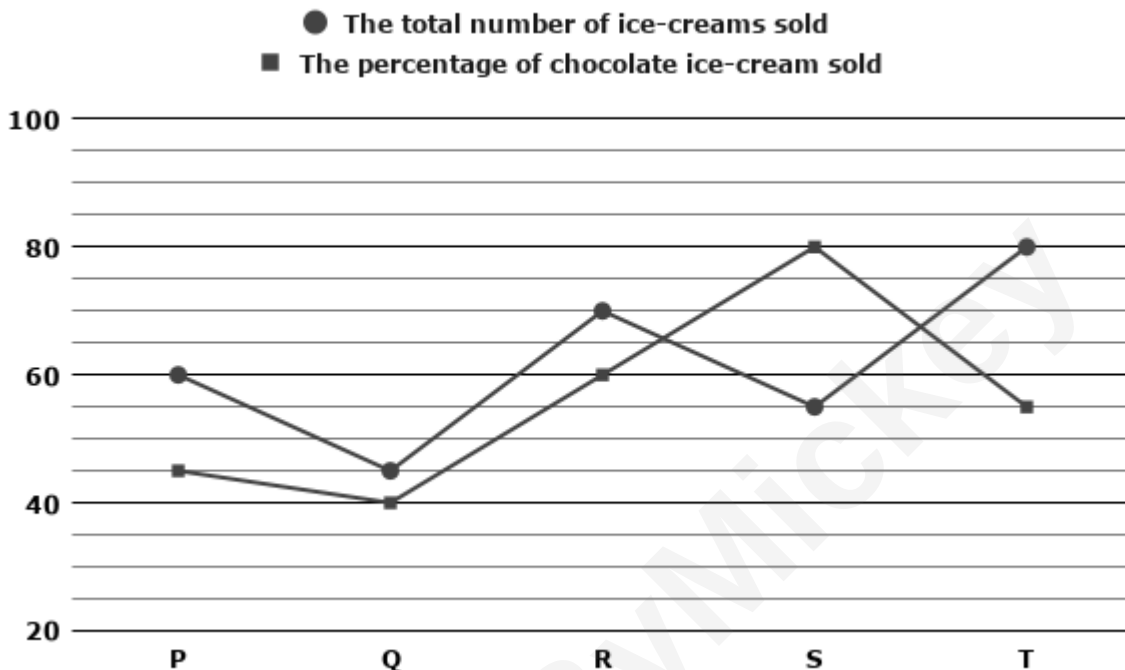


1. Questions

Study the following information carefully and answer the questions given below.

The line graph below shows the total number of ice-creams (vanilla and chocolates) sold by five different outlets along with the percentage of chocolate ice-creams sold by each outlet.



Note: The number of ice-creams sold = The number of vanilla ice-creams sold + The number of chocolate ice-creams sold.

The number of vanilla ice-creams sold by outlet P is increased by 33.33% and the number of chocolate ice-creams sold by outlet Q is increased by 11.11%, find the difference between the new number of vanilla ice-creams and the new number of chocolate ice-creams sold by outlets P and Q respectively.

- 24
- 20
- 40
- 80
- 70

2. Questions

What is the ratio of the total number of chocolate ice-creams sold by outlets Q and R together to the number of vanilla ice-creams sold by outlet T?

- 2 : 3
- 4 : 1

- c. 3 : 8
- d. 2 : 5
- e. 5 : 3

3. Questions

The total number of vanilla ice-creams sold by outlets Q and S together is what percentage of the total number of ice-creams sold by outlet T?

- a. 62.5%
- b. 47.5%
- c. 81.9%
- d. 66.6%
- e. 55.5%

4. Questions

If the average number of ice-creams sold by outlets Q and S is $2x$ and the difference between the total number of ice-creams sold by outlets R and P is $5y$. Find the value of $(x + 2y)$

- a. 26
- b. 31
- c. 28
- d. 29
- e. 23

5. Questions

The ratio of the number of chocolate ice-creams sold by outlet T to that by outlet U is 4 : 5, and the number of vanilla ice-creams sold by outlet U is 12 more than that by outlet R. Find the total number of ice-creams sold by outlet U.

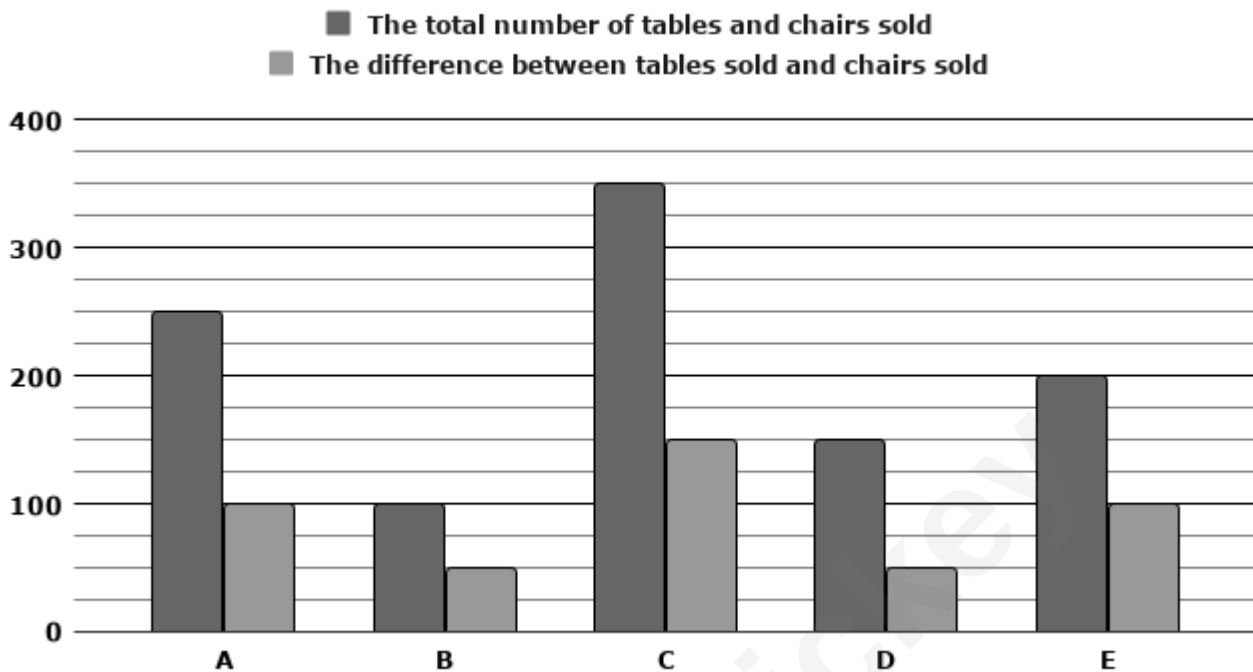
- a. 75
- b. 81
- c. 95
- d. 64
- e. 72

6. Questions

Study the following information carefully and answer the questions given below.

The bar graph given below shows the total number of tables and chairs sold by five different shops along

with the difference between the number of tables and chairs sold by each shop.



Notes:

(i). The total number of tables and chairs sold = The number of tables sold + The number of chairs sold.

(ii). The number of tables sold is greater than the number of chairs sold by each shop.

If the ratio between the number of plastic chairs and wooden chairs sold by shop D is 3 : 2, the number of plastic chairs sold by shop D is what percentage of the total number of chairs sold by shop B?

- a. 120%
- b. 110%
- c. 130%
- d. 140%
- e. 150%

7. Questions

What is the ratio between the average number of tables sold by shops C and E and the total number of chairs sold by shops A and E together?

- a. 4 : 1
- b. 1 : 3
- c. 8 : 5
- d. 5 : 9
- e. 3 : 4

8. Questions

If the total number of tables and chairs sold by shop X is 14.28% more than that by shop C. The number of chairs sold by shop X is $(14/15)$ of the number of chairs sold by shop A. Find the difference between the number of tables sold by shop X and the number of tables sold by shop B.

- a. 257
- b. 318
- c. 243
- d. 329
- e. 255

9. Questions

The difference between the number of tables sold by shops D and B is what percentage more or less than the difference between the number of chairs sold by shops C and D?

- a. 30% less
- b. 50% less
- c. 40% less
- d. 45% more
- e. 20% less

10. Questions

The total number of tables and chairs sold by shop D is what percentage more or less than that by shop E?

- a. 13.33% less
- b. 20% more
- c. 11.11% less
- d. 25% more
- e. 25% less

11. Questions

The average of 10 numbers is 45. When two new numbers are added, the average of all the numbers increases by 2.5. If the respective ratio of the two new numbers is 3 : 2, find the value of the greater number among the two new numbers.

- a. 84
- b. 75
- c. 62

d. 88

e. 72

12. Questions

Two trains, 280 meters and 320 meters in length, are running in opposite directions on parallel tracks. They cross each other completely in 12 seconds. If the speed of the first train is 72 km/h, what is the speed of the second train (in km/h)?

a. 90 km/h

b. 36 km/h

c. 54 km/h

d. 108 km/h

e. 72 km/h

13. Questions

A vessel contains 48 litres of a mixture of milk and water in the ratio 7 : 5 respectively. If 12 litres of milk and 5 litres of water is added to the mixture, what will be the new ratio of water to milk, respectively?

a. 8 : 3

b. 3 : 4

c. 5 : 8

d. 7 : 5

e. 9 : 7

14. Questions

Anubhav invests Rs. 12800 at 12.5% p.a simple interest for 2 years. He invests the interest earned in another scheme offering 10% p.a compound interest for 2 years. Find the compound interest earned from the second scheme.

a. Rs. 563

b. Rs. 672

c. Rs. 595

d. Rs. 664

e. Rs. 591

15. Questions

A shopkeeper marks the price of an article 60% above its cost price. He allows a discount of 25% on the marked price and earns a profit of Rs. 500 on the article. If he had given a discount of 35%

instead, what would have been his profit or loss percentage?

- a. 4%
- b. 3%
- c. 5%
- d. 6%
- e. 2%

16. Questions

A and B start a business by investing Rs. 22000 and Rs. 35000 respectively. After 8 months, A withdraws his entire investment. If the total profit earned at the end of the year is Rs. 44700, what is A's share of the profit?

- a. Rs. 13200
- b. Rs. 9800
- c. Rs. 10500
- d. Rs. 7400
- e. Rs. 12800

17. Questions

A can complete a piece of work in 19 days, while B can complete the same work in 38 days. They work together for 9 days, after which A leaves. In how many more days will B complete the remaining work?

- a. 9 days
- b. 11 days
- c. 7 days
- d. 13 days
- e. 5 days

18. Questions

A bag contains 14 balls out of which x are blue balls and the remaining balls are green balls. If two balls are drawn at random and the probability that both are blue balls is $\frac{4}{13}$, then find the number of green balls in the bag.

- a. 3
- b. 9
- c. 6
- d. 8

e. 5

19. Questions

A boat can travel 96 km downstream in 4 hours and the same distance upstream in 6 hours. If the speed of the boat in still water is increased by 25%, find the total time taken by the boat to cover 116 km downstream and 84 km upstream. (The speed of the stream remains constant).

- a. 9 hours
- b. 3 hours
- c. 5 hours
- d. 8 hours
- e. 7 hours

20. Questions

The ratio of the number of boys to girls in a class is 7 : 9. If the number of boys increases by 20% and the number of girls increases by 33.33%, find the new ratio of girls to boys in the class.

- a. 7 : 9
- b. 5 : 7
- c. 13 : 11
- d. 6 : 5
- e. 10 : 7

21. Questions

What value should come in the place of (?) in the following questions.

$(\frac{7}{5})$ of 75 + $(\frac{4}{9})$ of 81 = ? of $(\frac{5}{2})$

- a. 56.4
- b. 23.7
- c. 55.8
- d. 46.4
- e. 42.5

22. Questions

$648 \div 18 * 27 = 3^? * 36$

- a. 2
- b. 3

- c. 4
- d. 1
- e. 5

23. Questions

$66.67\% \text{ of } 720 - ? = 75\% \text{ of } 560$

- a. 50
- b. 20
- c. 60
- d. 40
- e. 30

24. Questions

$(36 - 72 \div 8) * 7 + 4 * ? = 25^2$

- a. 113
- b. 102
- c. 110
- d. 109
- e. 116

25. Questions

$72 \div 18 * 80 - ?\% \text{ of } 320 = 640 \div 4$

- a. 20
- b. 30
- c. 60
- d. 40
- e. 50

26. Questions**What approximate value will come in the place of (?) in the following questions?**

$35.01 * 24.98 + 749.7 \div 15.02 = ? * 4.98$

- a. 185
- b. 165

- c. 175
- d. 195
- e. 145

27. Questions

$$\sqrt{2400.99} - 13.99 * 4.01 + 18.05^2 = ?$$

- a. 320
- b. 317
- c. 324
- d. 311
- e. 329

28. Questions

$$65.05\% \text{ of } 1199.9 - 3.9/5.1 \text{ of } 349.9 = ? + 114.8$$

- a. 374
- b. 492
- c. 385
- d. 422
- e. 391

29. Questions

$$18.05^{1.89} + 13.99^{1.98} - 11.95 = ? + 160.05$$

- a. 334
- b. 312
- c. 329
- d. 348
- e. 376

30. Questions

$$124.95 / 4.99 * 11.98 + 44.95\% \text{ of } 599.98 = ?$$

- a. 530
- b. 420
- c. 510

d. 490

e. 570

31. Questions

What value will come in the place of (?) in the following number series?

49, 55, 67, ?, 109, 139

a. 85

b. 90

c. 75

d. 80

e. 95

32. Questions

7, 8, 18, 57, ?, 1165

a. 211

b. 232

c. 243

d. 223

e. 218

33. Questions

118, 137, 160, 189, 220, ?

a. 239

b. 241

c. 257

d. 231

e. 247

34. Questions

11550, 1050, ?, 30, 10, 5

a. 160

b. 130

c. 120

d. 150

e. 140

35. Questions

93, 107, 94, 108, ?, 109

a. 94

b. 92

c. 91

d. 90

e. 95

36. Questions

Find the wrong number in the following number series.

15, 21, 26, 33, 41, 50

a. 21

b. 15

c. 26

d. 41

e. 50

37. Questions

12, 7, 8, 14, 27, 68.5

a. 7

b. 14

c. 12

d. 68.5

e. 27

38. Questions

13, 21, 37, 61, 93, 132

a. 37

b. 13

c. 132

d. 61

e. 93

39. Questions**14, 50, 114, 215, 358, 554**

a. 114

b. 554

c. 14

d. 215

e. 50

40. Questions**11, 18, 32, 53, 81, 119**

a. 81

b. 53

c. 18

d. 11

e. 119

41. Questions

The following questions contain two equations as I and II. You have to solve both equations and determine the relationship between them and give answer as,

I). $x^2 - 43x + 460 = 0$

II). $y^2 - 39y + 380 = 0$

a. $x > y$ b. $x \geq y$ c. $x = y$ or relationship cannot be establishedd. $x < y$ e. $x \leq y$ **42. Questions**

I). $x^2 - 16x + 55 = 0$

II). $y^2 - 7y + 12 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

43. Questions

I). $x^2 - 26x + 168 = 0$

II). $y^2 - 31y + 238 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

44. Questions

I). $x^2 - 17x + 72 = 0$

II). $y^2 - 28y + 195 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

45. Questions

I). $x^2 - 11x + 30 = 0$

II). $y^2 - 10y + 21 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$

e. $x \leq y$

Explanations:

1. Questions

Outlet P:

The total number of ice-creams sold = 60

The number of chocolate ice-cream sold = $60 * 45\% = 27$

The number of vanilla ice-cream sold = $60 - 27 = 33$

Outlet Q:

The total number of ice-cream sold = 45

The number of chocolate ice-cream sold = $45 * 40\% = 18$

The number of vanilla ice-cream sold = $45 - 18 = 27$

Outlet R:

The total number of ice-creams sold = 70

The number of chocolate ice-cream sold = $70 * 60\% = 42$

The number of vanilla ice-cream sold = $70 - 42 = 28$

Outlet S:

The total number of ice-cream sold = 55

The number of chocolate ice-cream sold = $55 * 80\% = 44$

The number of vanilla ice-cream sold = $55 - 44 = 11$

Outlet T:

The total number of ice-cream sold = 80

The number of chocolate ice-cream sold = $80 * 55\% = 44$

The number of vanilla ice-cream sold = $80 - 44 = 36$

Outlet	The number of chocolate ice-cream sold	The number of vanilla ice-cream sold	The total number of ice-creams sold together
P	27	33	60
Q	18	27	45
R	42	28	70
S	44	11	55
T	44	36	80

Answer: A

The new number of vanilla ice-creams sold by outlet P = $33 \times \frac{4}{3} = 44$

The new number of chocolate ice-creams sold by outlet Q = $18 \times \frac{10}{9} = 20$

The required difference = $44 - 20 = 24$

Hence, the correct answer is option A.

2. Questions

Outlet P:

The total number of ice-creams sold = 60

The number of chocolate ice-cream sold = $60 \times 45\% = 27$

The number of vanilla ice-cream sold = $60 - 27 = 33$

Outlet Q:

The total number of ice-cream sold = 45

The number of chocolate ice-cream sold = $45 \times 40\% = 18$

The number of vanilla ice-cream sold = $45 - 18 = 27$

Outlet R:

The total number of ice-creams sold = 70

The number of chocolate ice-cream sold = $70 \times 60\% = 42$

The number of vanilla ice-cream sold = $70 - 42 = 28$

Outlet S:

The total number of ice-cream sold = 55

The number of chocolate ice-cream sold = $55 \times 80\% = 44$

The number of vanilla ice-cream sold = $55 - 44 = 11$

Outlet T:

The total number of ice-cream sold = 80

The number of chocolate ice-cream sold = $80 * 55\% = 44$

The number of vanilla ice-cream sold = $80 - 44 = 36$

Outlet	The number of chocolate ice-cream sold	The number of vanilla ice-cream sold	The total number of ice-creams sold together
P	27	33	60
Q	18	27	45
R	42	28	70
S	44	11	55
T	44	36	80

Answer: E

The total number of chocolate ice-creams sold by outlets Q and R together = $18 + 42 = 60$

The number of vanilla ice-creams sold by outlet T = 36

The required ratio = $60 : 36 = 5 : 3$

Hence, the correct answer is option E.

3. Questions

Outlet P:

The total number of ice-creams sold = 60

The number of chocolate ice-cream sold = $60 * 45\% = 27$

The number of vanilla ice-cream sold = $60 - 27 = 33$

Outlet Q:

The total number of ice-cream sold = 45

The number of chocolate ice-cream sold = $45 * 40\% = 18$

The number of vanilla ice-cream sold = $45 - 18 = 27$

Outlet R:

The total number of ice-creams sold = 70

The number of chocolate ice-cream sold = $70 * 60\% = 42$

The number of vanilla ice-cream sold = $70 - 42 = 28$

Outlet S:

The total number of ice-cream sold = 55

The number of chocolate ice-cream sold = $55 * 80\% = 44$

The number of vanilla ice-cream sold = $55 - 44 = 11$

Outlet T:

The total number of ice-cream sold = 80

The number of chocolate ice-cream sold = $80 * 55\% = 44$

The number of vanilla ice-cream sold = $80 - 44 = 36$

Outlet	The number of chocolate ice-cream sold	The number of vanilla ice-cream sold	The total number of ice-creams sold together
P	27	33	60
Q	18	27	45
R	42	28	70
S	44	11	55
T	44	36	80

Answer: B

The total number of vanilla ice-creams sold by outlets Q and S together = $27 + 11 = 38$

The total number of ice-creams sold by outlet T = 80

The required percentage = $38 * 100/80 = 47.5\%$

Hence, the correct answer is option B.

4. Questions

Outlet P:

The total number of ice-creams sold = 60

The number of chocolate ice-cream sold = $60 * 45\% = 27$

The number of vanilla ice-cream sold = $60 - 27 = 33$

Outlet Q:

The total number of ice-cream sold = 45

The number of chocolate ice-cream sold = $45 * 40\% = 18$

The number of vanilla ice-cream sold = $45 - 18 = 27$

Outlet R:

The total number of ice-creams sold = 70

The number of chocolate ice-cream sold = $70 * 60\% = 42$

The number of vanilla ice-cream sold = $70 - 42 = 28$

Outlet S:

The total number of ice-cream sold = 55

The number of chocolate ice-cream sold = $55 * 80\% = 44$

The number of vanilla ice-cream sold = $55 - 44 = 11$

Outlet T:

The total number of ice-cream sold = 80

The number of chocolate ice-cream sold = $80 * 55\% = 44$

The number of vanilla ice-cream sold = $80 - 44 = 36$

Outlet	The number of chocolate ice-cream sold	The number of vanilla ice-cream sold	The total number of ice-creams sold together
P	27	33	60
Q	18	27	45
R	42	28	70
S	44	11	55
T	44	36	80

Answer: D

The average number of ice-creams sold by outlets Q and S = $(45 + 55)/2 = 50$

So, $2x = 50$

$x = 25$

The difference between the ice-creams sold by outlets R and P = $70 - 60 = 10$

So, $5y = 10$

$$y = 2$$

$$\text{The required value} = 25 + 2 * 2 = 29$$

Hence, the correct answer is option D.

5. Questions

Outlet P:

$$\text{The total number of ice-creams sold} = 60$$

$$\text{The number of chocolate ice-cream sold} = 60 * 45\% = 27$$

$$\text{The number of vanilla ice-cream sold} = 60 - 27 = 33$$

Outlet Q:

$$\text{The total number of ice-cream sold} = 45$$

$$\text{The number of chocolate ice-cream sold} = 45 * 40\% = 18$$

$$\text{The number of vanilla ice-cream sold} = 45 - 18 = 27$$

Outlet R:

$$\text{The total number of ice-creams sold} = 70$$

$$\text{The number of chocolate ice-cream sold} = 70 * 60\% = 42$$

$$\text{The number of vanilla ice-cream sold} = 70 - 42 = 28$$

Outlet S:

$$\text{The total number of ice-cream sold} = 55$$

$$\text{The number of chocolate ice-cream sold} = 55 * 80\% = 44$$

$$\text{The number of vanilla ice-cream sold} = 55 - 44 = 11$$

Outlet T:

$$\text{The total number of ice-cream sold} = 80$$

$$\text{The number of chocolate ice-cream sold} = 80 * 55\% = 44$$

$$\text{The number of vanilla ice-cream sold} = 80 - 44 = 36$$

Outlet	The number of chocolate ice-cream sold	The number of vanilla ice-cream sold	The total number of ice-creams sold together
P	27	33	60
Q	18	27	45
R	42	28	70
S	44	11	55
T	44	36	80

Answer: C

The number of chocolate ice-creams sold by outlet U = $44 \times \frac{5}{4} = 55$

The number of vanilla ice-creams sold by outlet U = $28 + 12 = 40$

The total number of ice-creams sold by outlet U = $55 + 40 = 95$

Hence, the correct answer is option C.

6. Questions

Shop A:

The total number of tables and chairs sold by shop A = 250

The difference between tables and chairs sold by shop A = 100

The number of tables sold by shop A = $(250 + 100)/2 = 175$

The number of chairs sold by shop A = $(250 - 100)/2 = 75$

Shop B:

The total number of tables and chairs sold by shop B = 100

The difference between tables and chairs sold by shop B = 50

The number of tables sold by shop B = $(100 + 50)/2 = 75$

The number of chairs sold by shop B = $(100 - 50)/2 = 25$

Shop C:

The total number of tables and chairs sold by shop C = 350

The difference between tables and chairs sold by shop C = 150

The number of tables sold by shop C = $(350 + 150)/2 = 250$

The number of chairs sold by shop C = $(350 - 150)/2 = 100$

Shop D:

The total number of tables and chairs sold by shop D = 150

The difference between tables and chairs sold by shop D = 50

The number of tables sold by shop D = $(150 + 50)/2 = 100$

The number of chairs sold by shop D = $(150 - 50)/2 = 50$

Shop E:

The total number of tables and chairs sold by shop E = 200

The difference between tables and chairs sold by shop E = 100

The number of tables sold by shop E = $(200 + 100)/2 = 150$

The number of chairs sold by shop E = $(200 - 100)/2 = 50$

Shop	The number of tables sold	The number of chairs sold	The total number of tables and chairs sold together
A	175	75	250
B	75	25	100
C	250	100	350
D	100	50	150
E	150	50	200

Answer: A

The number of chairs sold by shop D = 50

The number of plastic chairs sold by shop D = $50 * 3/5 = 30$

The number of chairs sold by shop B = 25

The required percentage = $30 * 100/25 = 120\%$

Hence, the correct answer is option A.

7. Questions

Shop A:

The total number of tables and chairs sold by shop A = 250

The difference between tables and chairs sold by shop A = 100

The number of tables sold by shop A = $(250 + 100)/2 = 175$

The number of chairs sold by shop A = $(250 - 100)/2 = 75$

Shop B:

The total number of tables and chairs sold by shop B = 100

The difference between tables and chairs sold by shop B = 50

The number of tables sold by shop B = $(100 + 50)/2 = 75$

The number of chairs sold by shop B = $(100 - 50)/2 = 25$

Shop C:

The total number of tables and chairs sold by shop C = 350

The difference between tables and chairs sold by shop C = 150

The number of tables sold by shop C = $(350 + 150)/2 = 250$

The number of chairs sold by shop C = $(350 - 150)/2 = 100$

Shop D:

The total number of tables and chairs sold by shop D = 150

The difference between tables and chairs sold by shop D = 50

The number of tables sold by shop D = $(150 + 50)/2 = 100$

The number of chairs sold by shop D = $(150 - 50)/2 = 50$

Shop E:

The total number of tables and chairs sold by shop E = 200

The difference between tables and chairs sold by shop E = 100

The number of tables sold by shop E = $(200 + 100)/2 = 150$

The number of chairs sold by shop E = $(200 - 100)/2 = 50$

Shop	The number of tables sold	The number of chairs sold	The total number of tables and chairs sold together
A	175	75	250
B	75	25	100
C	250	100	350
D	100	50	150
E	150	50	200

Answer: C

The average number of tables sold by C and E = $(250 + 150)/2 = 200$

The total number of chairs sold by shops A and E together = $75 + 50 = 125$

The required ratio = $200 : 125 = 8 : 5$

Hence, the correct answer is option C.

8. Questions

Shop A:

The total number of tables and chairs sold by shop A = 250

The difference between tables and chairs sold by shop A = 100

The number of tables sold by shop A = $(250 + 100)/2 = 175$

The number of chairs sold by shop A = $(250 - 100)/2 = 75$

Shop B:

The total number of tables and chairs sold by shop B = 100

The difference between tables and chairs sold by shop B = 50

The number of tables sold by shop B = $(100 + 50)/2 = 75$

The number of chairs sold by shop B = $(100 - 50)/2 = 25$

Shop C:

The total number of tables and chairs sold by shop C = 350

The difference between tables and chairs sold by shop C = 150

The number of tables sold by shop C = $(350 + 150)/2 = 250$

The number of chairs sold by shop C = $(350 - 150)/2 = 100$

Shop D:

The total number of tables and chairs sold by shop D = 150

The difference between tables and chairs sold by shop D = 50

The number of tables sold by shop D = $(150 + 50)/2 = 100$

The number of chairs sold by shop D = $(150 - 50)/2 = 50$

Shop E:

The total number of tables and chairs sold by shop E = 200

The difference between tables and chairs sold by shop E = 100

The number of tables sold by shop E = $(200 + 100)/2 = 150$

The number of chairs sold by shop E = $(200 - 100)/2 = 50$

Shop	The number of tables sold	The number of chairs sold	The total number of tables and chairs sold together
A	175	75	250
B	75	25	100
C	250	100	350
D	100	50	150
E	150	50	200

Answer: E

The total number of tables and chairs sold by shop X = $(350 * 8/7) = 400$

The number of chairs sold by shop X = $75 * 14/15 = 70$

The number of tables sold by shop X = $400 - 70 = 330$

The number of tables sold by shop B = 75

The required difference = $330 - 75 = 255$

Hence, the correct answer is option E.

9. Questions

Shop A:

The total number of tables and chairs sold by shop A = 250

The difference between tables and chairs sold by shop A = 100

The number of tables sold by shop A = $(250 + 100)/2 = 175$

The number of chairs sold by shop A = $(250 - 100)/2 = 75$

Shop B:

The total number of tables and chairs sold by shop B = 100

The difference between tables and chairs sold by shop B = 50

The number of tables sold by shop B = $(100 + 50)/2 = 75$

The number of chairs sold by shop B = $(100 - 50)/2 = 25$

Shop C:

The total number of tables and chairs sold by shop C = 350

The difference between tables and chairs sold by shop C = 150

The number of tables sold by shop C = $(350 + 150)/2 = 250$

The number of chairs sold by shop C = $(350 - 150)/2 = 100$

Shop D:

The total number of tables and chairs sold by shop D = 150

The difference between tables and chairs sold by shop D = 50

The number of tables sold by shop D = $(150 + 50)/2 = 100$

The number of chairs sold by shop D = $(150 - 50)/2 = 50$

Shop E:

The total number of tables and chairs sold by shop E = 200

The difference between tables and chairs sold by shop E = 100

The number of tables sold by shop E = $(200 + 100)/2 = 150$

The number of chairs sold by shop E = $(200 - 100)/2 = 50$

Shop	The number of tables sold	The number of chairs sold	The total number of tables and chairs sold together
A	175	75	250
B	75	25	100
C	250	100	350
D	100	50	150
E	150	50	200

Answer: B

The difference between the tables sold by shops D and B = $100 - 75 = 25$

The difference between the chairs sold by shops C and D = $100 - 50 = 50$

The required percentage = $(50 - 25) * 100/50 = 50\%$ less

Hence, the correct answer is option B.

10. Questions

Shop A:

The total number of tables and chairs sold by shop A = 250

The difference between tables and chairs sold by shop A = 100

The number of tables sold by shop A = $(250 + 100)/2 = 175$

The number of chairs sold by shop A = $(250 - 100)/2 = 75$

Shop B:

The total number of tables and chairs sold by shop B = 100

The difference between tables and chairs sold by shop B = 50

The number of tables sold by shop B = $(100 + 50)/2 = 75$

The number of chairs sold by shop B = $(100 - 50)/2 = 25$

Shop C:

The total number of tables and chairs sold by shop C = 350

The difference between tables and chairs sold by shop C = 150

The number of tables sold by shop C = $(350 + 150)/2 = 250$

The number of chairs sold by shop C = $(350 - 150)/2 = 100$

Shop D:

The total number of tables and chairs sold by shop D = 150

The difference between tables and chairs sold by shop D = 50

The number of tables sold by shop D = $(150 + 50)/2 = 100$

The number of chairs sold by shop D = $(150 - 50)/2 = 50$

Shop E:

The total number of tables and chairs sold by shop E = 200

The difference between tables and chairs sold by shop E = 100

The number of tables sold by shop E = $(200 + 100)/2 = 150$

The number of chairs sold by shop E = $(200 - 100)/2 = 50$

Shop	The number of tables sold	The number of chairs sold	The total number of tables and chairs sold together
A	175	75	250
B	75	25	100
C	250	100	350
D	100	50	150
E	150	50	200

Answer: E

The total number of tables and chairs sold by shop D = 150

The total number of tables and chairs sold by shop E = 200

The required percentage = $(200 - 150) * 100/200 = 25\%$ less

Hence, the correct answer is option E.

11. Questions

Answer: E

The total of all 10 numbers = $45 * 10 = 450$

The new average of the all 12 numbers = $45 + 2.5 = 47.5$

The new total of all 12 numbers = $47.5 * 12 = 570$

The total of new two numbers = $570 - 450 = 120$

The greater number = $120 * 3/5 = 24 * 3 = 72$

Hence, the correct answer is option E.

12. Questions

Answer: D

The total distance travelled by the trains = $280 + 320 = 600$ m

The relative speed of the train in opposite direction = $600/12 = 50$ m/sec

The speed of the first train = $72 * 5/18 = 20$ m/sec

The speed of the seconds train = $50 - 20 = 30$ m/sec = $30 * 18/5 = 108$ km/h

Hence, the correct answer is option D.

13. Questions

Answer: C

The initial quantity of milk = $48 * \frac{7}{12} = 28$ litres

The initial quantity of water = $48 * \frac{5}{12} = 20$ litres

The new quantity of milk = $28 + 12 = 40$ litres

The new quantity of water = $20 + 5 = 25$ litres

The required ratio = $25 : 40 = 5 : 8$

Hence, the correct answer is option C.

14. Questions

Answer: B

The simple interest = $PRT/100$

The simple interest = $12800 * 12.5\% * 2 = \text{Rs. } 3200$

The effective rate of compound interest = $10 + 10 + 10 * \frac{10}{100} = 21\%$

The compound interest = $3200 * 21\% = \text{Rs. } 672$

Hence, the correct answer is option B.

15. Questions

Answer: A

Let the cost price of the article = $100a$

The marked price of the article = $100a * 160\% = 160a$

The selling price of the article = $160a * 75\% = 120a$

According to the question,

$$120a - 100a = 500$$

$$20a = 500$$

$$a = 25$$

The cost price of the article = 2500

The marked price of the article = $160 * 25 = 4000$

The new selling price of the article = $4000 * 65\% = 2600$

The new profit percentage = $(2600 - 2500) * \frac{100}{2500} = 4\%$ profit

Hence, the correct answer is option A.

16. Questions

Answer: A

The profit sharing ratio of A and B = $(22000 * 8) : (35000 * 12) = 44 : 105$

The profit share of A = $44a$

The profit share of B = $105a$

According to the question,

$$44a + 105a = 44700$$

$$149a = 44700$$

$$a = 300$$

The profit share of A = $44 * 300 = \text{Rs. } 13200$

Hence, the correct answer is option A.

17. Questions

Answer: B

Let the total work = 38 units [LCM of 19 and 38]

The efficiency of A = $38/19 = 2$ units per day

The efficiency of B = $38/38 = 1$ unit per day

The total efficiency of A and B = $2 + 1 = 3$ units per day

The work done in 9 days = $3 * 9 = 27$ units

The remaining work = $38 - 27 = 11$ units

The work done by B = $11/1 = 11$ days

Hence, the correct answer is option B.

18. Questions

Answer: C

The total number of balls = 14

The number of blue balls = x

The number of green balls = $14 - x$

According to the question,

$$[x(x - 1)]/14 * 13 = 4 / 13$$

$$x(x - 1) = 56$$

$$x^2 - x - 56 = 0$$

$$x^2 - 8x + 7x - 56 = 0$$

$$x(x - 8) + 7(x - 8) = 0$$

$$(x - 8)(x + 7) = 0$$

$$x = 8 \text{ or } -7$$

The number of blue balls = 8

The number of green balls = $14 - 8 = 6$

Hence, the correct answer is option C.

19. Questions

Answer: D

The downstream speed of the boat = $96/4 = 24$ km/h

The upstream speed of the boat = $96/6 = 16$ km/h

The speed of the boat in still water = $(24 + 16)/2 = 20$ km/h

The speed of the stream = $(24 - 16)/2 = 4$ km/h

The new speed of the boat in still water = $20 * 125/100 = 25$ km/h

The new downstream speed of the boat = $(25 + 4) = 29$ km/h

The new upstream speed of the boat = $25 - 4 = 21$ km/h

The required answer = $(116/29) + (84/21) = 8$ hours

Hence, the correct answer is option D.

20. Questions

Answer: E

Let the number of boys in the class = $7a$

The number of girls in the class = $9a$

The new number of boys in the class = $7a * 120/100 = 8.4a$

The new number of girls in the class = $9a * (4/3) = 12a$

The required ratio = $12a : 8.4a = 10 : 7$

Hence, the correct answer is option E.

21. Questions

Answer: A

$(7/5)$ of $75 + (4/9)$ of $81 = ?$ of $(5/2)$

$105 + 36 = ? * (5/2)$

$141 = ? * (5/2)$

$56.4 = ?$

Hence, the correct answer is option A.

22. Questions

Answer: B

$$648 \div 18 * 27 = 3^? * 36$$

$$36 * 27 = 3^? * 36$$

$$27 = 3^?$$

$$3^3 = 3^?$$

$$? = 3$$

Hence, the correct answer is option B.

23. Questions

Answer: C

$$66.67\% \text{ of } 720 - ? = 75\% \text{ of } 560$$

$$(2/3) * 720 - ? = (3/4) * 560$$

$$480 - ? = 420$$

$$480 - ? = 420$$

$$? = 60$$

Hence, the correct answer is option C.

24. Questions

Answer: D

$$(36 - 72 \div 8) * 7 + 4 * ? = 25^2$$

$$(36 - 9) * 7 + 4 * ? = 625$$

$$27 * 7 + 4 * ? = 625$$

$$189 + 4 * ? = 625$$

$$436 = 4 * ?$$

$$109 = ?$$

Hence, the correct answer is option D.

25. Questions

Answer: E

$$72 \div 18 * 80 - ?\% \text{ of } 320 = 640 \div 4$$

$$320 - ?\% * 320 = 160$$

$$160 = ?\% \text{ of } 320$$

$$160 = (? / 100) * 320$$

$$50 = ?$$

Hence, the correct answer is option E.

26. Questions

Answer: A

$$35.01 * 24.98 + 749.7 \div 15.02 = ? * 4.98$$

$$35 * 25 + 750 \div 15 = ? * 5$$

$$875 + 50 = ? * 5$$

$$925 = ? * 5$$

$$185 = ?$$

Hence, the correct answer is option A.

27. Questions

Answer: B

$$\sqrt{2400.99} - 13.99 * 4.01 + 18.05^2 = ?$$

$$\sqrt{2401} - 14 * 4 + 18^2 = ?$$

$$49 - 56 + 324 = ?$$

$$373 - 56 = ?$$

$$317 = ?$$

Hence, the correct answer is option B.

28. Questions

Answer: C

$$65.05\% \text{ of } 1199.9 - 3.9/5.1 \text{ of } 349.9 = ? + 114.8$$

$$65\% \text{ of } 1200 - (4/5) \text{ of } 350 = ? + 115$$

$$780 - 280 = ? + 115$$

$$500 = ? + 115$$

$$? = 385$$

Hence, the correct answer is option C.

29. Questions

Answer: D

$$18.05^{1.89} + 13.99^{1.98} - 11.95 = ? + 160.05$$

$$18^2 + 14^2 - 12 = ? + 160$$

$$324 + 196 - 12 = ? + 160$$

$$520 - 12 = ? + 160$$

$$508 = ? + 160$$

$$348 = ?$$

Hence, the correct answer is option D.

30. Questions

Answer: E

$$124.95 / 4.99 * 11.98 + 44.95\% \text{ of } 599.98 = ?$$

$$125 / 5 * 12 + 45\% \text{ of } 600 = ?$$

$$300 + 270 = ?$$

$$570 = ?$$

Hence, the correct answer is option E.

31. Questions

Answer: A

$$49 + 6 = 55$$

$$55 + 12 = 67$$

$$67 + 18 = \mathbf{85}$$

$$85 + 24 = 109$$

$$109 + 30 = 139$$

Missing number: 85

Hence, the correct answer is option A.

32. Questions

Answer: B

$$7 * 1 + 1 = 8$$

$$8 * 2 + 2 = 18$$

$$18 * 3 + 3 = 57$$

$$57 * 4 + 4 = \mathbf{232}$$

$$232 * 5 + 5 = 1165$$

Missing number: 232

Hence, the correct answer is option B.

33. Questions

Answer: C

$$118 + 19 = 137$$

$$137 + 23 = 160$$

$$160 + 29 = 189$$

$$189 + 31 = 220$$

$$220 + 37 = \mathbf{257}$$

Missing number: 257

Hence, the correct answer is option C.

34. Questions

Answer: D

$$11550 \div 11 = 1050$$

$$1050 \div 7 = \mathbf{150}$$

$$150 \div 5 = 30$$

$$30 \div 3 = 10$$

$$10 \div 2 = 5$$

Missing number: 150

Hence, the correct answer is option D.

35. Questions

Answer: E

$$93 + 14 = 107$$

$$107 - 13 = 94$$

$$94 + 14 = 108$$

$$108 - 13 = \mathbf{95}$$

$$95 + 14 = 109$$

Missing number: 95

Hence, the correct answer is option E.

36. Questions

Answer: A

$$15 + 5 = \mathbf{20}$$

$$20 + 6 = 26$$

$$26 + 7 = 33$$

$$33 + 8 = 41$$

$$41 + 9 = 50$$

Wrong number: 21

Correct number: 20

Hence, the correct answer is option A.

37. Questions

Answer: B

$$12 * 0.5 + 1 = 7$$

$$7 * 1 + 1 = 8$$

$$8 * 1.5 + 1 = \mathbf{13}$$

$$13 * 2 + 1 = 27$$

$$27 * 2.5 + 1 = 68.5$$

Wrong number: 14

Correct number: 13

Hence, the correct answer is option B.

38. Questions

Answer: C

$$13 + 8 = 21$$

$$21 + 16 = 37$$

$$37 + 24 = 61$$

$$61 + 32 = 93$$

$$93 + 40 = \mathbf{133}$$

Wrong number: 132

Correct number: 133

Hence, the correct answer is option C.

39. Questions

Answer: D

$$14 + 6^2 = 50$$

$$50 + 8^2 = 114$$

$$114 + 10^2 = \mathbf{214}$$

$$214 + 12^2 = 358$$

$$358 + 14^2 = 554$$

Wrong number: 215

Correct number: 214

Hence, the correct answer is option D.

40. Questions

Answer: E

$$11 + 7 = 18$$

$$18 + 14 = 32$$

$$32 + 21 = 53$$

$$53 + 28 = 81$$

$$81 + 35 = \mathbf{116}$$

Wrong number: 119

Correct number: 116

Hence, the correct answer is option E.

41. Questions

Answer: B

$$\text{I). } x^2 - 43x + 460 = 0$$

$$x^2 - 23x - 20x + 460 = 0$$

$$x(x - 23) - 20(x - 23) = 0$$

$$(x - 23)(x - 20) = 0$$

$$x = 23 \text{ or } 20$$

$$\text{II). } y^2 - 39y + 380 = 0$$

$$y^2 - 20y - 19y + 380 = 0$$

$$y(y - 20) - 19(y - 20) = 0$$

$$(y - 19)(y - 20) = 0$$

$$y = 19 \text{ or } 20$$

So, $x \geq y$

Hence, the correct answer is option B.

42. Questions

Answer: A

I). $x^2 - 16x + 55 = 0$

$$x^2 - 11x - 5x + 55 = 0$$

$$x(x - 11) - 5(x - 11) = 0$$

$$(x - 5)(x - 11) = 0$$

$$x = 5 \text{ or } 11$$

II). $y^2 - 7y + 12 = 0$

$$y^2 - 4y - 3y + 12 = 0$$

$$y(y - 4) - 3(y - 4) = 0$$

$$(y - 3)(y - 4) = 0$$

$$y = 3 \text{ or } 4$$

$$\text{So, } x > y$$

Hence, the correct answer is option A.

43. Questions

Answer: E

I). $x^2 - 26x + 168 = 0$

$$x^2 - 14x - 12x + 168 = 0$$

$$x(x - 14) - 12(x - 14) = 0$$

$$(x - 14)(x - 12) = 0$$

$$x = 14 \text{ or } 12$$

II). $y^2 - 31y + 238 = 0$

$$y^2 - 17y - 14y + 238 = 0$$

$$y(y - 17) - 14(y - 17) = 0$$

$$(y - 14)(y - 17) = 0$$

$$y = 14 \text{ or } 17$$

$$\text{So, } x \leq y$$

Hence, the correct answer is option E.

44. Questions

Answer: D

I). $x^2 - 17x + 72 = 0$

$$x^2 - 9x - 8x + 72 = 0$$

$$x(x - 9) - 8(x - 9) = 0$$

$$(x - 8)(x - 9) = 0$$

$$x = 8 \text{ or } 9$$

II). $y^2 - 28y + 195 = 0$

$$y^2 - 13y - 15y + 195 = 0$$

$$y(y - 13) - 15(y - 13) = 0$$

$$(y - 15)(y - 13) = 0$$

$$y = 15 \text{ or } 13$$

$$\text{So, } x < y$$

Hence, the correct answer is option D.

45. Questions

Answer: C

I). $x^2 - 11x + 30 = 0$

$$x^2 - 6x - 5x + 30 = 0$$

$$x(x - 6) - 5(x - 6) = 0$$

$$(x - 5)(x - 6) = 0$$

$$x = 5 \text{ or } 6$$

II). $y^2 - 10y + 21 = 0$

$$y^2 - 7y - 3y + 21 = 0$$

$$y(y - 7) - 3(y - 7) = 0$$

$$(y - 3)(y - 7) = 0$$

$$y = 3 \text{ or } 7$$

So, $x = y$ or relationship cannot be established.

Hence, the correct answer is option C.